Jeongbin Sohn (손정빈)

123, Cheomdangwagi-ro
Buk-gu, Gwangju, Republic of Korea
sjb010409@gm.gist.ac.kr, sjb010409@gmail.com
LinkedIn: www.linkedin.com/in/jeongbin-sohn-robotics
Website: caesar0409.github.io

RESEARCH INTERESTS

Vision-based Navigation, Robot Perception, Computer Vision, Spacecraft Autonomy

EDUCATION

GIST (Gwangju Institute of Science and Technology) Mar. 2020 - Feb. 2027
B.S. Mechanical and Robotics Engineering Major /
Electrical Engineering and Computer Science Minor
Cumulative GPA: 4.38/4.5

University of California, Berkeley Jan. 2026 - May 2026
Study Abroad Program

Boston University Jul. 2021 - Aug. 2021
Student Exchange Program

Gwangju Science Academy for the Gifted (GSA) Mar. 2017 - Feb. 2020
High School Graduation

RESEARCH EXPERIENCE

UC Berkeley Embodied Dexterity Group (EDG) Jan. 2026 - Jun. 2026
Student Intern
Supervisor: Prof. Hannah Stuart

- *Analyzed recorded videos of individuals with C5–C7 spinal cord injuries demonstrating the dorsal grasper.*

GIST Machine Perception and Intelligence Lab (MPIL) Jan. 2025 - Current
Student Intern
Supervisor: Prof. Pyojin Kim

- *Developed SIGNN++, a graph neural network-based star identification framework robust to terrestrial noise such as light pollution, clouds, and motion blur; achieved a 98.5% solved rate on real night-sky images, compared with 79.1% for the strongest existing method.*
- *Built a real-world evaluation setup using a Rocket Lab ST-RT star tracker as the attitude reference.*
- *Adapted star identification algorithms to lunar crater identification in collaboration with the Korea Institute of Energy Technology (KENTECH).*

GIST Bio-Robotics Lab Jul. 2024 - Dec. 2024
Student Intern
Supervisor: Prof. Jongho Lee

- *Studied the Robot Operating System (ROS) and path-planning algorithms.*

KITECH (Korea Institute of Industrial Technology) Aug. 2023 - Feb. 2024
Research Assistant
Supervisor: Dr. Hojae Lee

- *Ported existing code to Python and implemented a domain-change algorithm for indoor localization.*

PUBLICATIONS

1. Conference Presentations
   - **Sohn***, Kim*, Kim, and Lee 2026 IROS SRW, "*Craters as Constellations: Adapting Star Identification to Lunar Crater Identification*", Accepted for Poster

Presentation (*equal contribution)
- Park, **Sohn**, and Lee 2025 ICROS, "*A Low-Cost Celestial Compass Under Low-Light Featureless Environment*", Accepted for Poster Presentation

2. Under Review
- **Sohn**, Lee, Yen, Gwon, Han, Hepburn-Dickins, Coltin, Soussan, and Kim 2027 ICRA, "*SIGNN++: Robust Graph Neural Star Identification on Earth*"

AWARDS & FELLOWSHIPS

Study Abroad Program Scholarship Jan. 2026 - May 2026
Awarded 26,000 USD to eight students selected for the Study Abroad Program (SAP) at UC Berkeley.

Korea National Science and Engineering Scholarship Mar. 2020 - Feb. 2027
Full-tuition scholarship for eight semesters, funded by the Korea Student Aid Foundation, Ministry of Education. (국가이공계장학금)

Korea-U.S. Exchange Program for Advanced Fields Jan. 2026 - May 2026
Merit-based scholarship of 9,000 USD awarded by the Korea Institute for Advancement of Technology (KIAT) to 113 undergraduate students in South Korea.

Woonhae Scholarship (운해장학금) Mar. 2025 - Feb. 2026
Merit-based scholarship of 2,780 USD per semester for one year, awarded by the Woonhae Foundation to students with a GPA above 4.0/4.5.

U.S. Army Commendation Medal Jun. 2023
GIST Mech-Eng Internship Presentation Award (2nd place) Jun. 2021

TEACHING EXPERIENCE

Teaching Assistant Sep. 2024 - Dec. 2024
GIST GS2001 Multivariable Calculus and Applications

TECHNOLOGY SKILLS

Programming Languages:
Proficient: Python (PyTorch, PyTorch Geometric, NumPy)
Working knowledge: C++, C, MATLAB
Others: ROS, LabVIEW, Siemens NX, COMSOL Multiphysics, OpenFOAM

LANGUAGE PROFICIENCY

Korean (native), English (fluent[1]), Japanese (basic knowledge)

OTHER ACTIVITIES

GIST Swimming Club Leader Mar. 2025 - Aug. 2025
Led training sessions and social events, fostering teamwork among club members.

ELITE Club Founding Member Jul. 2025 - Dec. 2025
Founding member of ELITE (English for Learning, Inquiry, Thought, and Expression), an English study club supervised by Prof. Ellis Lee; planned club activities and led group studies.

KSEA Member Feb. 2026 - May 2026
Member of the Korean-American Scientists & Engineers Association (KSEA) at UC Berkeley; participated in the academic journal club.

Language Exchange Program Feb. 2026 - May 2026
Learned English and taught Korean with a UC Berkeley student.

[1] 1.5 years as a KATUSA (Korean Augmentation To the United States Army) at a U.S. Army base, and exchange programs at Boston University and UC Berkeley